Class:11 (Protein Structure Prediction with AlphaFold)

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Background

We ran 3 AlphaFold2 predictions,

- HIV-PR_Monomer
- HIV-PR_dimer
- My find-a-gene protein sequence

Here we analyze the resulting models and make some diagnostic plots that inform on model reliability

1. HIV-PR Monomer

```
list.files(path= "HIV_monomer_94b5b_2/" ,
           full.names = F,
           pattern = ".pdb")
```

```
[1] "HIV_monomer_94b5b_2_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.pdb"
[2] "HIV_monomer_94b5b_2_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.pdb"
[3] "HIV_monomer_94b5b_2_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb"
[4] "HIV_monomer_94b5b_2_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000.pdb"
[5] "HIV_monomer_94b5b_2_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000.pdb"
```

```
pdb_files <- list.files(path= "HIV_monomer_94b5b_2/" ,
                        full.names = T,
                        pattern = ".pdb")
```

```
library(bio3d)

pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
HIV_monomer_94b5b_2//HIV_monomer_94b5b_2_unrelaxed_rank_001_alphafold2_ptm_model_4_seed_000.p
HIV_monomer_94b5b_2//HIV_monomer_94b5b_2_unrelaxed_rank_002_alphafold2_ptm_model_5_seed_000.p
HIV_monomer_94b5b_2//HIV_monomer_94b5b_2_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.p
HIV_monomer_94b5b_2//HIV_monomer_94b5b_2_unrelaxed_rank_004_alphafold2_ptm_model_1_seed_000.p
HIV_monomer_94b5b_2//HIV_monomer_94b5b_2_unrelaxed_rank_005_alphafold2_ptm_model_2_seed_000.p
.....

Extracting sequences

pdb/seq: 1   name: HIV_monomer_94b5b_2//HIV_monomer_94b5b_2_unrelaxed_rank_001_alphafold2_ptm
pdb/seq: 2   name: HIV_monomer_94b5b_2//HIV_monomer_94b5b_2_unrelaxed_rank_002_alphafold2_ptm
pdb/seq: 3   name: HIV_monomer_94b5b_2//HIV_monomer_94b5b_2_unrelaxed_rank_003_alphafold2_ptm
pdb/seq: 4   name: HIV_monomer_94b5b_2//HIV_monomer_94b5b_2_unrelaxed_rank_004_alphafold2_ptm
pdb/seq: 5   name: HIV_monomer_94b5b_2//HIV_monomer_94b5b_2_unrelaxed_rank_005_alphafold2_ptm
```

```
pdbs
```

```
                                 1        .         .         .         .         50
[Truncated_Name:1]HIV_monome     PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]HIV_monome     PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]HIV_monome     PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]HIV_monome     PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]HIV_monome     PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                                 **************************************************
                                 1        .         .         .         .         50

                                51        .         .         .         .        99
[Truncated_Name:1]HIV_monome     GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]HIV_monome     GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]HIV_monome     GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]HIV_monome     GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]HIV_monome     GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                                 **************************************************
                                51        .         .         .         .        99

Call:
```

```
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 99 position columns (99 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

RMSD

Calculate the RMSD of all models to each other (i.e. pairwise RMSD)

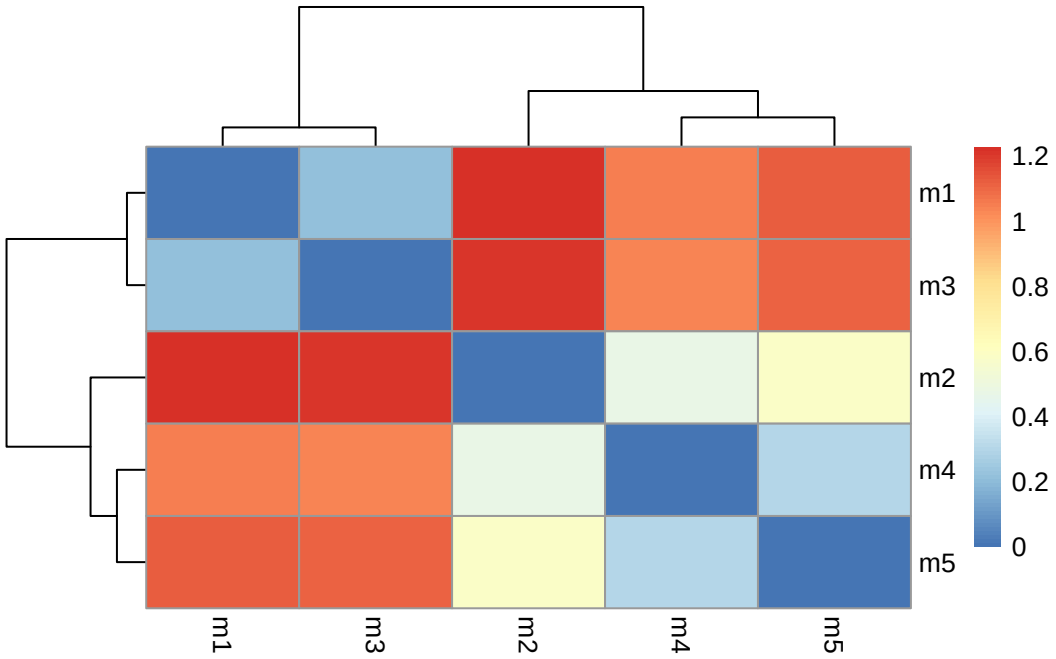
```
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 99 non NA positions
```

and a heatmap display

```
library(pheatmap)

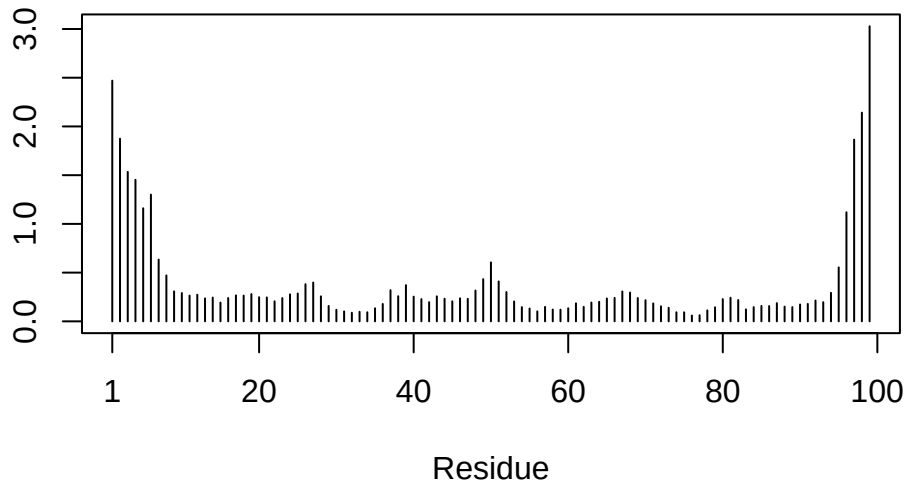
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



RMSF

Calculate RMSF - how far apart each amino-acid is accross all 5 AlphaFold models

```
rf <- rmsf(pdbs$xyz)
plot.bio3d(rf)
```



Residue Conservation

```
aln_file <- list.files(path="HIV_monomer_94b5b_2/",
                       pattern=".a3m$",
                       full.names = TRUE)
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
```

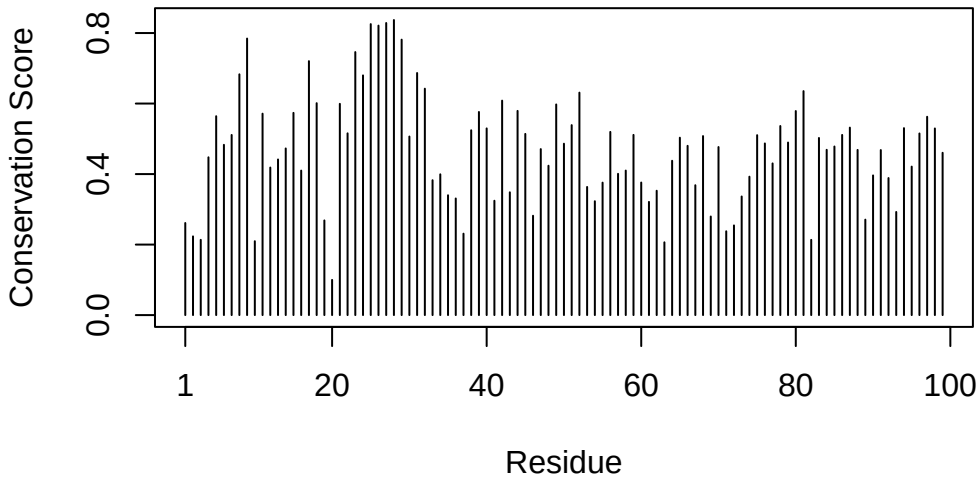
```
dim(aln$ali)
```

```
[1] 5397  132
```

```
sim <- conserv(aln)
```

```
plotb3(sim[1:99], sse=trim.pdbs(pdbs, chain="A"),
       ylab="Conservation Score")
```

```
Warning in plotb3(sim[1:99], sse = trim.pdbs(pdbs, chain = "A"), ylab =
"Conservation Score"): Length of input 'sse' does not equal the length of input
'x'; Ignoring 'sse'
```




```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

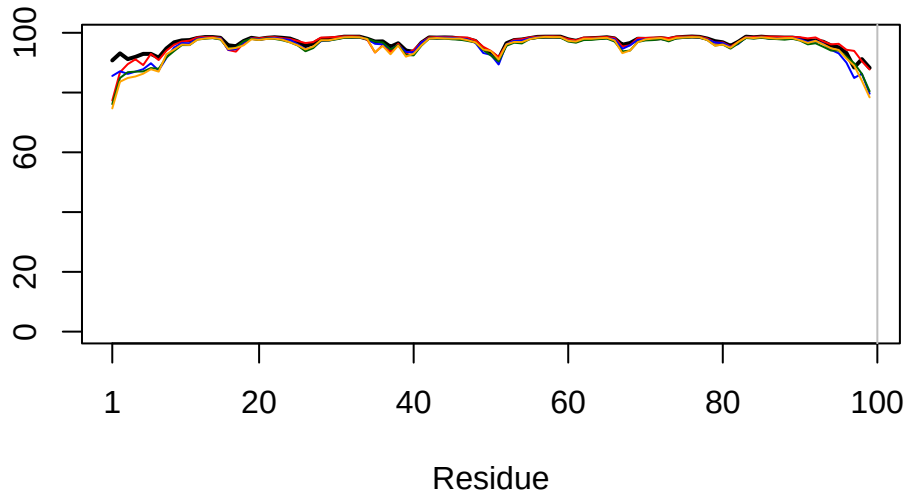
pLDDT & PAE score plots

The pLDDT scores are stroed in the PDB models "B-factor" col and in a sepereate JSON format file in the results directory. We can use either:

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdbs)
```

```
Warning in plotb3(pdbs$b[1, ], typ = "l", lwd = 2, sse = pdbs): Length of input
'sse' does not equal the length of input 'x'; Ignoring 'sse'
```

```
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



2. Hive-PR Dimer

```
list.files(path="HIV_Dimer_23119/",

full.names = F,
pattern=".pdb")
```

```
[1] "HIV_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "HIV_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "HIV_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "HIV_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "HIV_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
pdb_files <- list.files(path="HIV_Dimer_23119/",

full.names = T,
pattern="*.pdb")
# Print our PDB file names
basename(pdb_files)
```

```
[1] "HIV_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "HIV_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "HIV_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "HIV_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "HIV_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
HIV_Dimer_23119//HIV_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.p
HIV_Dimer_23119//HIV_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.p
HIV_Dimer_23119//HIV_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.p
HIV_Dimer_23119//HIV_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.p
HIV_Dimer_23119//HIV_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.p
.....

Extracting sequences

pdb/seq: 1   name: HIV_Dimer_23119//HIV_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3
pdb/seq: 2   name: HIV_Dimer_23119//HIV_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3
pdb/seq: 3   name: HIV_Dimer_23119//HIV_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3
pdb/seq: 4   name: HIV_Dimer_23119//HIV_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3
pdb/seq: 5   name: HIV_Dimer_23119//HIV_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3
```

```
pdbs
```

```
                              1        .         .         .         .        50
[Truncated_Name:1]HIV_dimer_  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]HIV_dimer_  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]HIV_dimer_  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]HIV_dimer_  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]HIV_dimer_  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                              **************************************************
                              1        .         .         .         .        50

                             51        .         .         .         .       100
[Truncated_Name:1]HIV_dimer_  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]HIV_dimer_  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]HIV_dimer_  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]HIV_dimer_  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]HIV_dimer_  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                              **************************************************
                             51        .         .         .         .       100

                            101        .         .         .         .       150
[Truncated_Name:1]HIV_dimer_  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]HIV_dimer_  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]HIV_dimer_  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]HIV_dimer_  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]HIV_dimer_  QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                              **************************************************
                            101        .         .         .         .       150

                            151        .         .         .         .     198
[Truncated_Name:1]HIV_dimer_  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]HIV_dimer_  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]HIV_dimer_  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]HIV_dimer_  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]HIV_dimer_  GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                              *********************************************
                            151        .         .         .         .     198

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
```

```
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

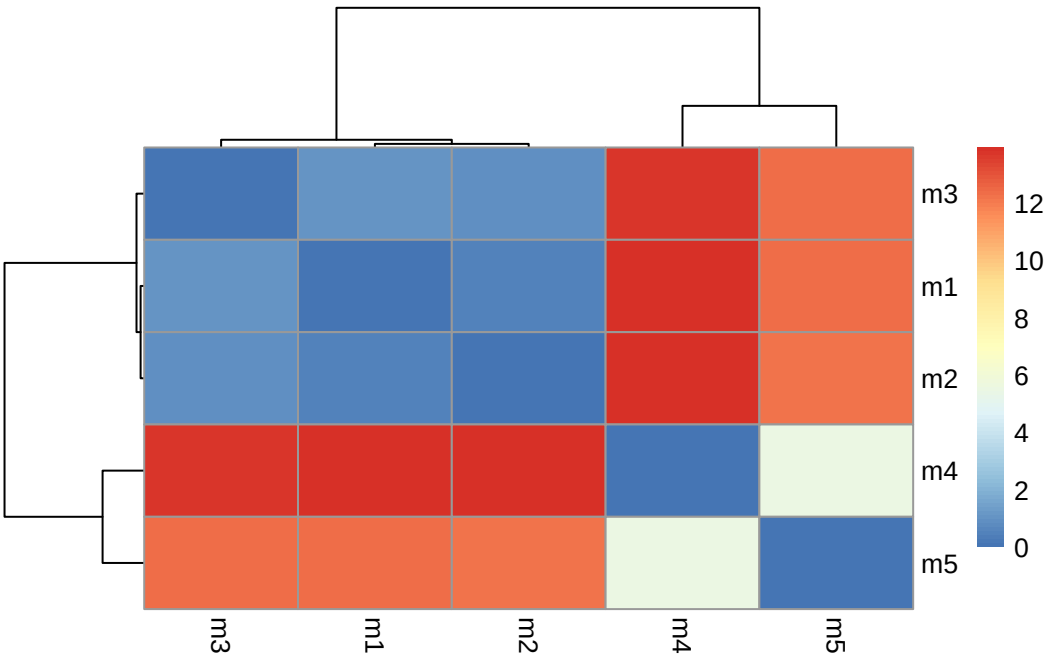
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

RMSD

```
rd <- rmsd(pdbs, fit=T)
```

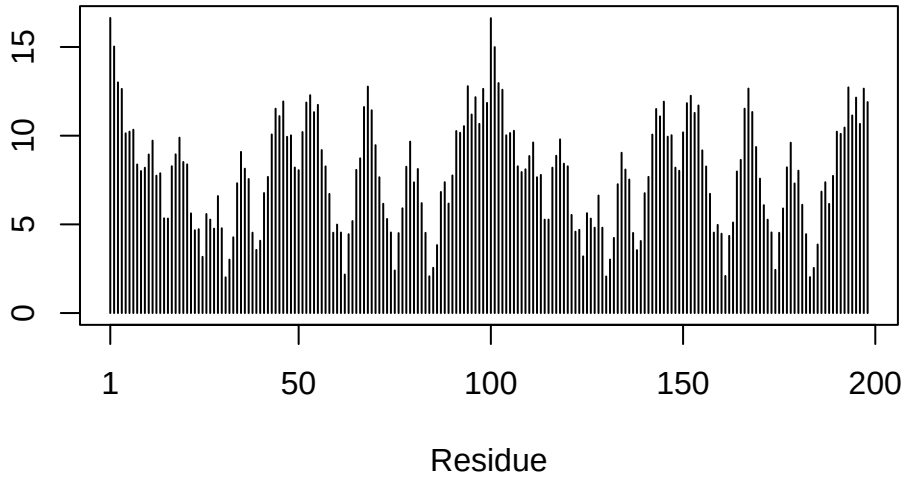
```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```
library(pheatmap)
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



RMSF

```
rf <- rmsf(pdbs$xyz)
plot.bio3d(rf)
```



Residue Conservation

```
aln_file <- list.files(path="HIV_Dimer_23119/",
pattern=".a3m$",
full.name=TRUE)

aln_file
```

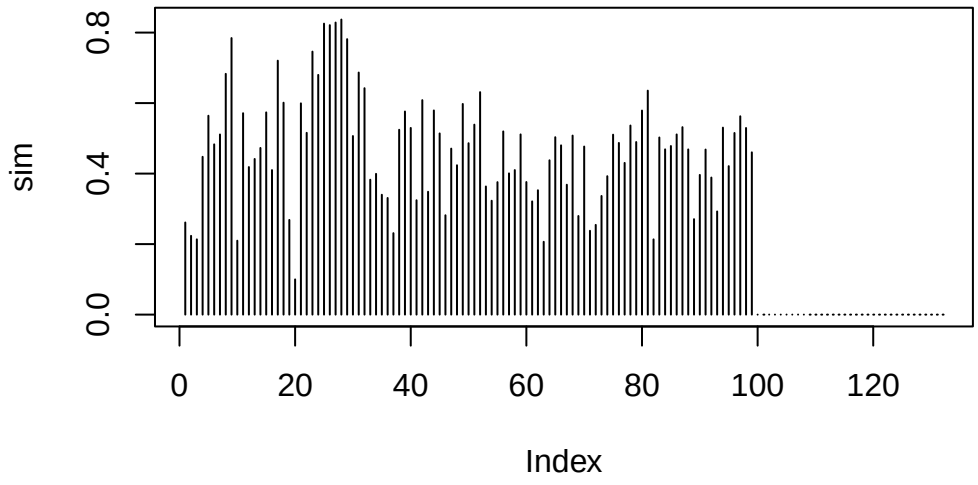
```
[1] "HIV_Dimer_23119//HIV_dimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper=TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
sim <- conserv(aln)
```

```
plot(sim, typ="h")
```




```
con <- consensus(aln, cutoff=0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

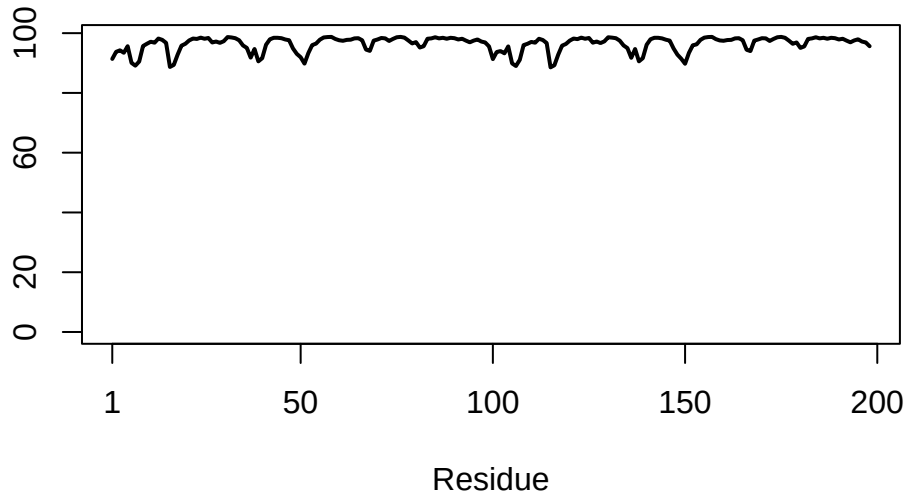
```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(rep(sim[1:99], 2), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_dimer_conserv.pdb")
```

pLDDT & PAE score plots

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=m1.pdb)
```

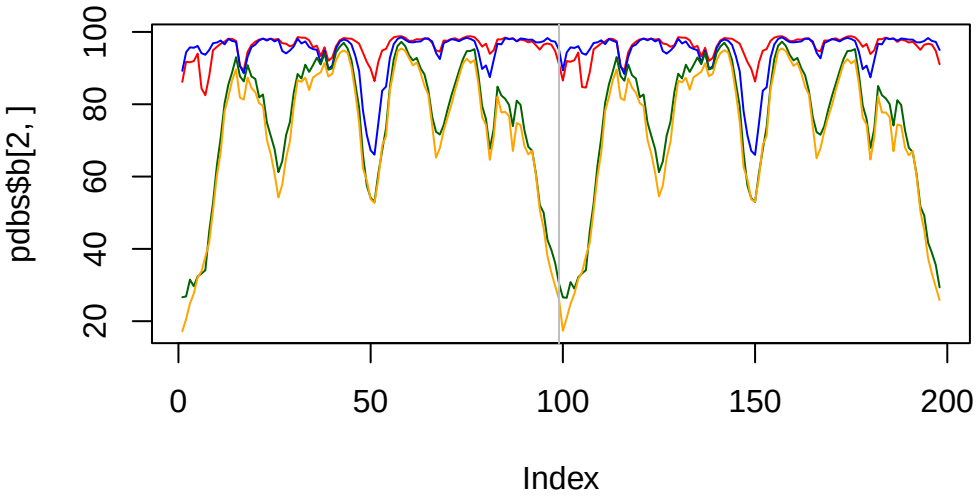
```
Warning in pdb2sse(sse): No helix and sheet defined in input 'sse' PDB object:
try using dssp()
```

```
Warning in plotb3(pdbs$b[1, ], typ = "l", lwd = 2, sse = m1.pdb): Length of
input 'sse' does not equal the length of input 'x'; Ignoring 'sse'
```




```
# Initialize the plot first using the first dataset range
plot(pdbs$b[2, ], type="l", col="red", ylim=c(min(pdbs$b, na.rm=TRUE), max(pdbs$b, na.rm=TRU

# Then add the remaining lines
points(pdbs$b[3, ], type="l", col="blue")
points(pdbs$b[4, ], type="l", col="darkgreen")
points(pdbs$b[5, ], type="l", col="orange")
abline(v=99, col="gray")
```



3. My Find-A-Gene Sequence

```
list.files(path="test_e1404",

full.names = F,
pattern=".pdb")
```

```
[1] "test_e1404_unrelaxed_rank_001_alphafold2_ptm_model_5_seed_000.pdb"
[2] "test_e1404_unrelaxed_rank_002_alphafold2_ptm_model_4_seed_000.pdb"
[3] "test_e1404_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb"
[4] "test_e1404_unrelaxed_rank_004_alphafold2_ptm_model_2_seed_000.pdb"
[5] "test_e1404_unrelaxed_rank_005_alphafold2_ptm_model_1_seed_000.pdb"
```

```
pdb_files <- list.files(path="test_e1404",

full.names = T,
pattern="*.pdb")
# Print our PDB file names
basename(pdb_files)
```

```
[1] "test_e1404_unrelaxed_rank_001_alphafold2_ptm_model_5_seed_000.pdb"
[2] "test_e1404_unrelaxed_rank_002_alphafold2_ptm_model_4_seed_000.pdb"
[3] "test_e1404_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb"
[4] "test_e1404_unrelaxed_rank_004_alphafold2_ptm_model_2_seed_000.pdb"
[5] "test_e1404_unrelaxed_rank_005_alphafold2_ptm_model_1_seed_000.pdb"
```

```
library(bio3d)
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
test_e1404/test_e1404_unrelaxed_rank_001_alphafold2_ptm_model_5_seed_000.pdb
test_e1404/test_e1404_unrelaxed_rank_002_alphafold2_ptm_model_4_seed_000.pdb
test_e1404/test_e1404_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.pdb
test_e1404/test_e1404_unrelaxed_rank_004_alphafold2_ptm_model_2_seed_000.pdb
test_e1404/test_e1404_unrelaxed_rank_005_alphafold2_ptm_model_1_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: test_e1404/test_e1404_unrelaxed_rank_001_alphafold2_ptm_model_5_seed_000.p
pdb/seq: 2   name: test_e1404/test_e1404_unrelaxed_rank_002_alphafold2_ptm_model_4_seed_000.p
pdb/seq: 3   name: test_e1404/test_e1404_unrelaxed_rank_003_alphafold2_ptm_model_3_seed_000.p
pdb/seq: 4   name: test_e1404/test_e1404_unrelaxed_rank_004_alphafold2_ptm_model_2_seed_000.p
pdb/seq: 5   name: test_e1404/test_e1404_unrelaxed_rank_005_alphafold2_ptm_model_1_seed_000.p
```

```
pdbs
```

```
                                 1        .         .         .         .         50
[Truncated_Name:1]test_e1404   QYYAGWTDKMTGKVIPVDGQIFSYTRHEPVGVCGAIIPWNYPLLMISWKV
[Truncated_Name:2]test_e1404   QYYAGWTDKMTGKVIPVDGQIFSYTRHEPVGVCGAIIPWNYPLLMISWKV
[Truncated_Name:3]test_e1404   QYYAGWTDKMTGKVIPVDGQIFSYTRHEPVGVCGAIIPWNYPLLMISWKV
[Truncated_Name:4]test_e1404   QYYAGWTDKMTGKVIPVDGQIFSYTRHEPVGVCGAIIPWNYPLLMISWKV
[Truncated_Name:5]test_e1404   QYYAGWTDKMTGKVIPVDGQIFSYTRHEPVGVCGAIIPWNYPLLMISWKV
                               **************************************************
                               1        .         .         .         .         50

                              51        .         .         .         .         100
[Truncated_Name:1]test_e1404   GPALACGCTLVLKPAEQTPLTALYFAALVKEAGFPPGVVNVIPGYGPTAG
[Truncated_Name:2]test_e1404   GPALACGCTLVLKPAEQTPLTALYFAALVKEAGFPPGVVNVIPGYGPTAG
[Truncated_Name:3]test_e1404   GPALACGCTLVLKPAEQTPLTALYFAALVKEAGFPPGVVNVIPGYGPTAG
[Truncated_Name:4]test_e1404   GPALACGCTLVLKPAEQTPLTALYFAALVKEAGFPPGVVNVIPGYGPTAG
```

```
[Truncated_Name:5]test_e1404   GPALACGCTLVLKPAEQTPLTALYFAALVKEAGFPPGVVNVIPGYGPTAG
                               **************************************************
                              51        .         .         .         .        100

                             101        .         .         .         .        150
[Truncated_Name:1]test_e1404   TAIAEHMDINKVAFTGSLEVGRLIMQASGKSNLKRVTLELGGKSPFVIFA
[Truncated_Name:2]test_e1404   TAIAEHMDINKVAFTGSLEVGRLIMQASGKSNLKRVTLELGGKSPFVIFA
[Truncated_Name:3]test_e1404   TAIAEHMDINKVAFTGSLEVGRLIMQASGKSNLKRVTLELGGKSPFVIFA
[Truncated_Name:4]test_e1404   TAIAEHMDINKVAFTGSLEVGRLIMQASGKSNLKRVTLELGGKSPFVIFA
[Truncated_Name:5]test_e1404   TAIAEHMDINKVAFTGSLEVGRLIMQASGKSNLKRVTLELGGKSPFVIFA
                               **************************************************
                             101        .         .         .         .        150

                             151        .         .         .         .        200
[Truncated_Name:1]test_e1404   DADLDEAVMWAHSGIMNNHGQNCCAGSRTFVEAPIYDKFVKRCKELAEKR
[Truncated_Name:2]test_e1404   DADLDEAVMWAHSGIMNNHGQNCCAGSRTFVEAPIYDKFVKRCKELAEKR
[Truncated_Name:3]test_e1404   DADLDEAVMWAHSGIMNNHGQNCCAGSRTFVEAPIYDKFVKRCKELAEKR
[Truncated_Name:4]test_e1404   DADLDEAVMWAHSGIMNNHGQNCCAGSRTFVEAPIYDKFVKRCKELAEKR
[Truncated_Name:5]test_e1404   DADLDEAVMWAHSGIMNNHGQNCCAGSRTFVEAPIYDKFVKRCKELAEKR
                               **************************************************
                             151        .         .         .         .        200

                             201        .         .         .         .        250
[Truncated_Name:1]test_e1404   KVGDPWEASTQQGPQIDEEQFNRIKDLIDSGVKEGAKLECGGSRYGDKGF
[Truncated_Name:2]test_e1404   KVGDPWEASTQQGPQIDEEQFNRIKDLIDSGVKEGAKLECGGSRYGDKGF
[Truncated_Name:3]test_e1404   KVGDPWEASTQQGPQIDEEQFNRIKDLIDSGVKEGAKLECGGSRYGDKGF
[Truncated_Name:4]test_e1404   KVGDPWEASTQQGPQIDEEQFNRIKDLIDSGVKEGAKLECGGSRYGDKGF
[Truncated_Name:5]test_e1404   KVGDPWEASTQQGPQIDEEQFNRIKDLIDSGVKEGAKLECGGSRYGDKGF
                               **************************************************
                             201        .         .         .         .        250

                             251        .         .         .         .        300
[Truncated_Name:1]test_e1404   FVQPTVFSNVQDNMRIAKEEIFGPVQSILKFTDIEEVIERAKQTPYGLAA
[Truncated_Name:2]test_e1404   FVQPTVFSNVQDNMRIAKEEIFGPVQSILKFTDIEEVIERAKQTPYGLAA
[Truncated_Name:3]test_e1404   FVQPTVFSNVQDNMRIAKEEIFGPVQSILKFTDIEEVIERAKQTPYGLAA
[Truncated_Name:4]test_e1404   FVQPTVFSNVQDNMRIAKEEIFGPVQSILKFTDIEEVIERAKQTPYGLAA
[Truncated_Name:5]test_e1404   FVQPTVFSNVQDNMRIAKEEIFGPVQSILKFTDIEEVIERAKQTPYGLAA
                               **************************************************
                             251        .         .         .         .        300

                             301302
[Truncated_Name:1]test_e1404   GV
[Truncated_Name:2]test_e1404   GV
```

```
[Truncated_Name:3]test_e1404   GV
[Truncated_Name:4]test_e1404   GV
[Truncated_Name:5]test_e1404   GV
                               **
                            301302

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 302 position columns (302 non-gap, 0 gap)

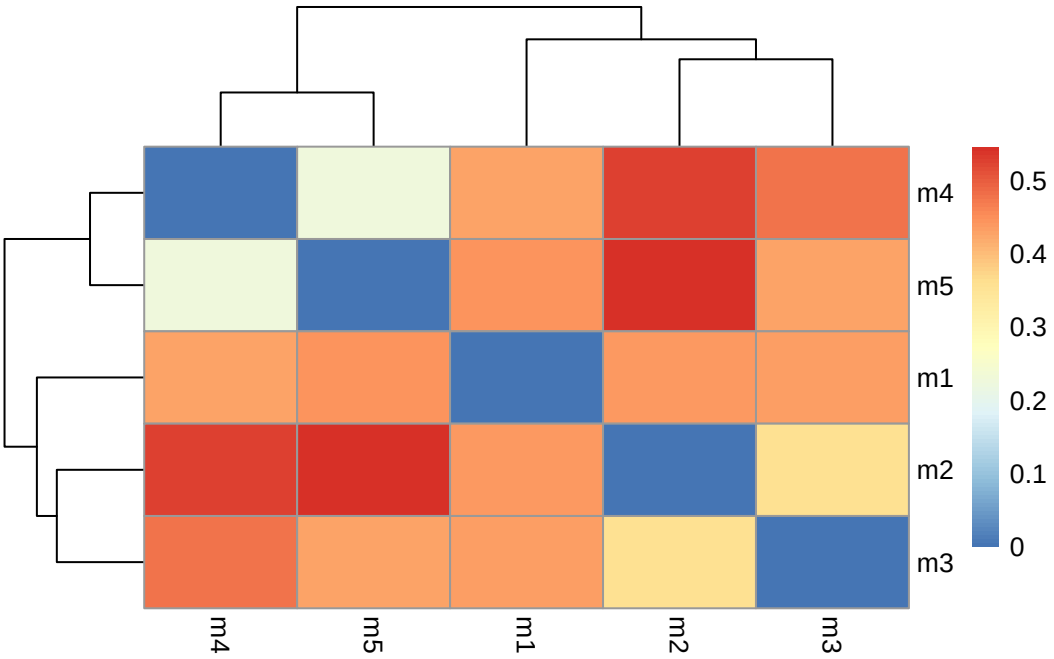
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

RMSD

```
rd <- rmsd(pdbs, fit=T)
```

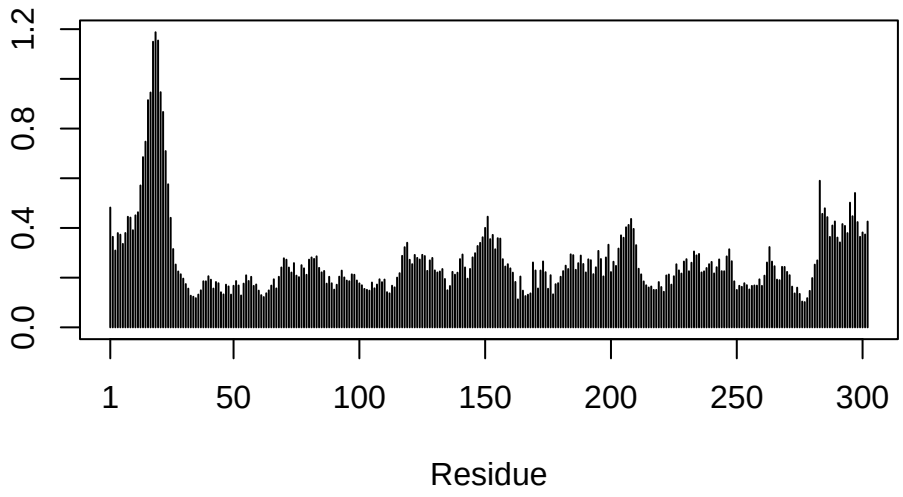
```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 302 non NA positions
```

```
library(pheatmap)
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



RMSF

```
rf <- rmsf(pdbs$xyz)
plot.bio3d(rf)
```



Residue Conservation

```
aln_file <- list.files(path= "test_e1404",

pattern=".a3m$",
full.name=TRUE)

aln_file
```

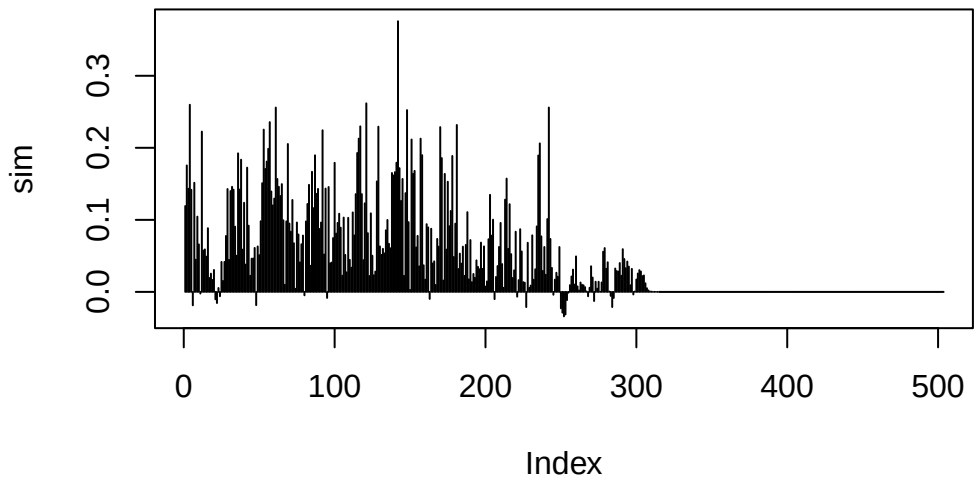
```
[1] "test_e1404/test_e1404.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper=TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
```

```
sim <- conserv(aln)
```

```
plot(sim, typ="h")
```




```
con <- consensus(aln, cutoff=0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[145] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[163] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[181] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[199] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[217] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[235] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[253] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[271] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[289] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[307] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[325] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
```

```
[343] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[361] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[379] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[397] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[415] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[433] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[451] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[469] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[487] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
```

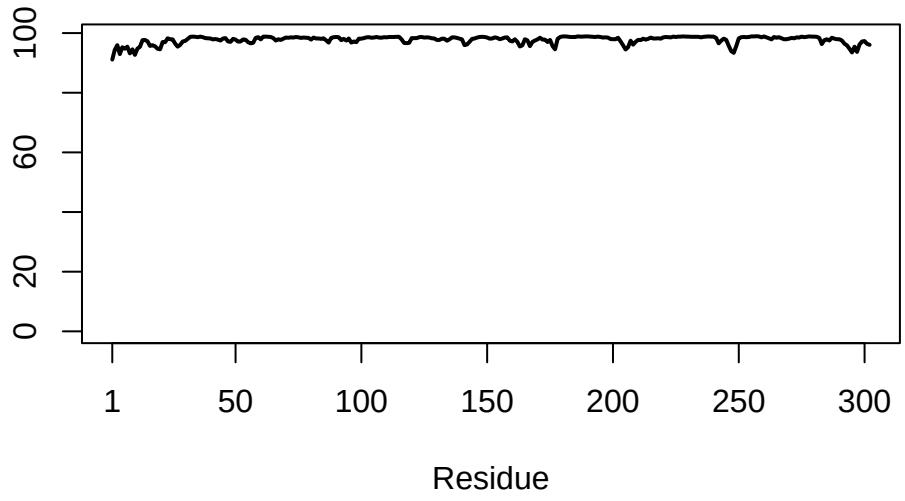
```
m1.pdb <- read.pdb(pdb_files[1])
n_res <- length(unique(m1.pdb$atom$resno))
occ <- vec2resno(sim[1:n_res], m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

pLDDT & PAE score plots

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=m1.pdb)
```

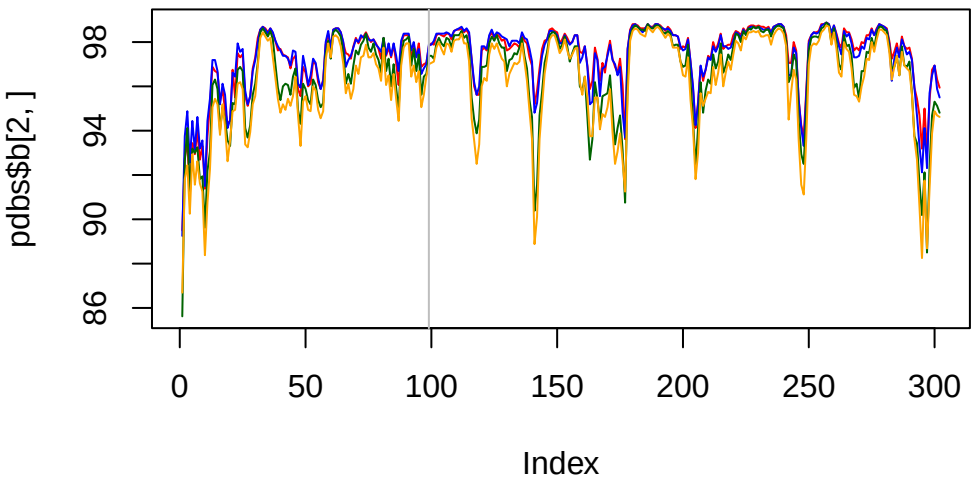
```
Warning in pdb2sse(sse): No helix and sheet defined in input 'sse' PDB object:
try using dssp()
```

```
Warning in plotb3(pdbs$b[1, ], typ = "l", lwd = 2, sse = m1.pdb): Length of
input 'sse' does not equal the length of input 'x'; Ignoring 'sse'
```




```
plot(pdbs$b[2, ], type="l", col="red", ylim=c(min(pdbs$b, na.rm=TRUE), max(pdbs$b, na.rm=TRU

# Then add the remaining lines
points(pdbs$b[3, ], type="l", col="blue")
points(pdbs$b[4, ], type="l", col="darkgreen")
points(pdbs$b[5, ], type="l", col="orange")
abline(v=99, col="gray")
```


pdbs$b[2,]
86
90
94
98
0
50
100
150
200
250
300
Index